

Evaluation of automated-trading strategies using an artificial market

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ABSTRACT

The purpose of this study was to extend the simulation of artificial markets to the practical application. We constructed an artificial-market system with an interface with the automated-trading strategies. Then, using an artificial-market simulation, we conducted two types of evaluations of automated-trading strategies that could participate in a Kaburobo competition. (1) Our system evaluated the risks and returns of the strategies in various market environments. An evaluation using the artificial market was able to provide better information than a conventional evaluation using a back test. (2) Our system could also test the market impact of automated-trading strategies. Our results revealed that the market impact of the strategies may not only depend on their rule content but also on the way they are combined with other strategies.

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1. Introduction

An artificial market is an agent-based model of a financial market. Studies on artificial markets have attained some successes in market analysis in recent years. Many of them were, however, too simple and abstract to be a useful model of actual markets. Especially trading rules of agents were very restricted and simplified in comparison to trading strategies used in actual markets. The purpose of this study was to extend the simulation of artificial markets to the practical application. We constructed a new platform for artificial-market simulation using a standardized protocol of electronic trading in actual markets. Then, it was used for the evaluation of automated-trading strategies.

New computer technology called automated trading has attracted a great deal of attention in financial markets in recent years [17,27]. It is also known as algorithmic trading or robo trading.¹ Automated trading is a generic name for the technology of dealing in assets, such as stocks, bonds, and currencies, automatically by computers. This involves a set of rules to determine the timing, price, and quantity of an order. For example, the golden cross rule is that a signal to buy is made when a security's short-term moving average (such as 20-day

moving average) breaks above its long-term moving average (such as 50-day moving average). When a rule is satisfied, an automated-trading system automatically places an order on the market through an electronic-broking system. Some automated-trading systems combine different trading rules and use machine learning to find optimal parameters for those rules. Automated trading is increasingly becoming the executive tool of choice for traders to accomplish operational efficiency. By the end of 2010, the Aite Group² estimates that approximately 53% of all trading in equities will be done through automated trading [2]. This rapid growth is based on enhanced computer performance and the improved information environment in financial markets. Despite such a rapid spread of automated trading, methods of assessing it have not yet been established. Therefore, traders cannot know how they expect it to perform or its risks until they actually use it in the market. Moreover, automated trading can no longer be ignored as its spread has had an impact on markets. It is particularly important to know whether the spread of certain types of trading software will threaten the stability of financial markets.

We propose the evaluation of automated trading using artificial-market simulation as an approach to solving these problems. We evaluated automated-trading strategies that

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¹ In some definitions, automatically determining how trade is conducted is called algorithmic trading, and automatic determination including the timing, price, and quantity of an order is called automated trading [3].

² The Aite Group is one of the most outstanding research and advisory firms focused on finance and technology [1]. It published many important reports about the impact of technology on the financial services industry.

participated in a competition of automated trading called Kaburobo³ [15,14,26]. The evaluation using artificial-market simulation had more advantages than a back test using past market data. First, the artificial market can generate an ensemble of price paths with their possibility under some market conditions, including what had not occurred in actual markets. Thus, it could be used to evaluate automated trading in various market environments. Then, an evaluation result using simulation could be robust than a result by a back test. Second, as the artificial market could be used to simulate a feedback loop between the market and automated-trading strategies, it could be used to evaluate the market impact of automated trading. Thus, the artificial market could be a test-bed for machine learning of automated-trading strategies.

The outline of this paper is as follows. First, Kaburobo is introduced and the specifications for automated-trading strategies that participated in a competition are explained. Second, we describe the framework of the system we used to evaluate automated-trading strategies using an artificial market. Third, we used the artificial market to estimate how well automated trading performed and the risks it involved. The results of the evaluation were compared with those from a back test using past market data. Finally, we evaluated the market impact of automated-trading strategies using the artificial market.

2. Kaburobo: competition between automated-trading strategies

Kaburobo is a programming contest for stock-trading software agents, held by the Trade Science Corporation [25]. The aim of this competition is dissemination of automated trading, and the sponsor is planning commercialization of high-quality trading strategies that are selected from a couple of thousand agents in this contest. Participants have various backgrounds; computer engineers, professional dealers, academic researchers, undergraduate students, and so on. This leads to the variety of trading strategies in the contest.

2.1. Rules of Kaburobo contest

Trading rules of the Kaburobo contest is based on those in the Tokyo Stock Exchange [15,20,26].

Initial assets: All software agents individually invest 50 m yen (about 0.5 m dollars) virtually on the Tokyo Stock Exchange, during certain periods such as those lasting several months. The agents determine their trading behavior autonomously, and the participants cannot operate nor change their agents during the competition.

Stocks: The stocks that agents can trade are all stocks included in Nikkei-225 index and 300 stocks with larger trading volumes. They cover almost all major stocks listed on the first section of the Tokyo Stock Exchange.

Order method: During the contest, all agents determine the kinds of stocks, price, and quantity of orders automatically using market data provided by the Kaburobo contest server. They also select a type of orders; a market order or a limit order. A market order is an order to buy or sell at a market price, and a limit order is an order at the price that a agent mentioned. Short-selling is allowed, but simultaneous short-selling and spot-buying of the same stock are prohibited. It is because such trading behavior fakes larger liquidity of the stock than actual.

Timing of orders: Agents can place their orders twice a day; ends of morning session and afternoon session. At each session,

they can place multiple orders to buy and/or sell a number of stocks, within amounts of their capitals.

Order flow: The orders of agents are sent to the Kaburobo contest server. A function of connection to the server can be implemented by a Kaburobo software development kit (SDK), which the sponsor provide. The server does not place the orders in the actual market, but stores the order data. Thus, orders in the Kaburobo contest are virtual, and cannot affect actual stock prices.

Evaluation: After the contest is over, the total profits and losses of the virtual investments are calculated by taking expenses, such as commissions and interest rates, into consideration. Commissions are 0.1% of executed volumes, and interests are charged for short-selling. The agents are ranked by quantitatively evaluating their profits and losses in the contest, the results of a back test using past market data, and qualitative evaluations by professional traders.

2.2. Overlearning problem of Kaburobo agents

The Kaburobo competition has been held every year since 2004 [19]. More than 6000 participants and 8000 agents participated in the competition held in 2006 (Table 1). Various automatic-trading strategies have participated in the competition. Some agents have implemented various methods of economic forecasting and technical analysis, and combined them. Other agents have used artificial intelligence technology such as reinforcement learning, support vector machines, or genetic algorithms. These agents have been written in Java with one of two methods [20]. The first is where a agent is developed in detail using a software development kit (Kaburobo SDK), which the management office prepared. Agents with complicated strategies can be freely created with this method. The second is where tools are used to create agents by setting them up simply by using a graphical user interface (GUI). This method is used to create agents that combine investment indexes, such as the moving average and relative strength index (RSI).

The top 10 agents in the latest competition invested 50 m yen (about 0.5 m dollars) in an actual market for half a year (from February to August, 2007) as an out-of-sample test [15]. As a result, the agents were not able to obtain results as good as those from a back test (Table 2). The average profit (annual rate) of the 10 agents was about three times as much as the Tokyo stock price index (TOPIX) in a back test using past data from 1997 to 2006. However, they only yielded the same return as TOPIX in the out-of-sample test. This fact raised doubts on the reliability of the evaluation by a back test. Overlearning on given data set may have caused these different results between the back and the out-of-sample tests. Since the competition was ranked using past data, parameters of agents were adjusted for these. This overlearning problem also occurred in U-Mart, another contest of software trading agents [22]: Some types of agents could attain very high performance with one data set, but they went bankrupt with the other data sets. Thus, the back test using a given data set is not enough for testing automated-trading strategies.

Table 1
History of Kaburobo competition.

	2004	2005	2006
Investment period	1 month	3 months	4 months
Number of stocks	40	100	500
Number of participants	2405	2735	6264
Number of agents	3200	4681	8367
Best return (annual rate) (%)	332	828	56.84

³ “Kabu” means “stock” in Japanese.

Table 2

Average profit (annual rate) of the top 10 programs.

	Back test (%)										Out-of-sample test (%)
	1997	1998	1999	2000	2001	2002	2003	2004	2005	2006	2007
Kaburobo	1.2	16.1	44.1	36.6	10.8	11.5	21.5	12.9	17.0	9.5	−9.8
TOPIX	−20.1	−7.5	58.4	−25.5	−19.6	−18.3	23.8	10.2	43.5	1.9	−9.4

3. System for evaluating automated-trading strategies using artificial market

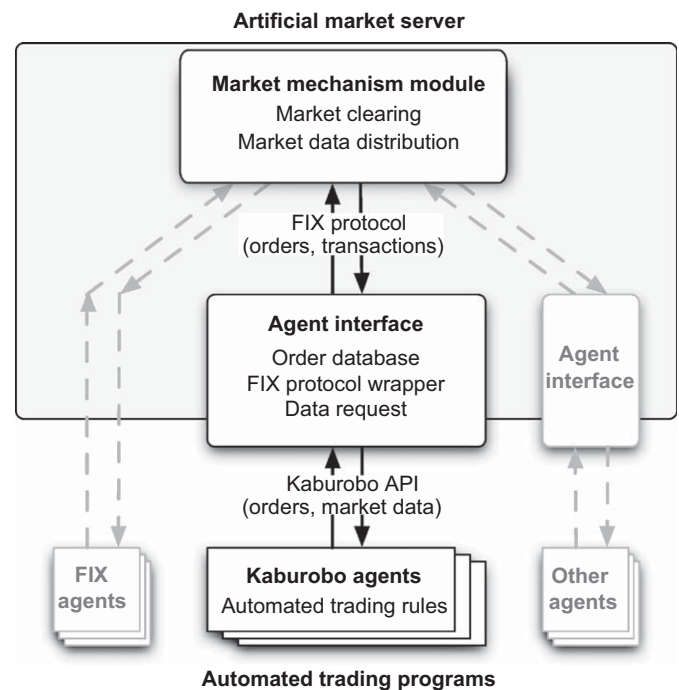
Studies on artificial markets belong to the agent-based computational economics (ACE) approach [13,23]. Their purpose is to analyze market phenomena from the viewpoint of the interaction among heterogeneous financial agents, and they have attained some successes. They have especially found new mechanisms for market phenomena that previous models could not adequately explain. For example, the Santa Fe Artificial Stock Market could simulate financial bubbles with agents using genetic algorithms [4,21]. Using an artificial market, Lux explained the fat-tailed distribution of financial returns, one of stylized facts of financial markets [18]. Artificial markets were also used to test existing economic theories such as efficient market hypotheses [7]. As a market design using an artificial market, Darley and Outkin examined the effect of the tick-size reduction in the Nasdaq market [9]. From theoretical to empirical studies, various results are obtained by artificial markets in these 10 years [5,6,8].

Despite many successful results in market analysis, results of the practical application of artificial markets are still limited. One of the main reasons is that trading rules of agents are unlike those of actual dealers. In some artificial market models, the design of agents is very simplified for the ease of analysis. For example, agents only can choose their rules from two simple trading rules; *chartists* (one-step extrapolation) or *fundamentalists* (regression to a given fundamental price level). In other models, agents must create their trading rules from scratch using machine learning methods. Acquired rules are sometimes too complicated for humans to understand. In order to overcome this problem, we constructed a new platform for artificial-market simulation using a standardized protocol of electronic trading in actual markets. Then, trading strategies in an actual market can participate in our artificial market. Thus, our artificial market could be used to the evaluation of automated-trading strategies.

3.1. Framework for simulation of artificial markets

The two crucial components of an artificial market are agents and the market mechanism. Agents, i.e., computer programs that play the role of virtual dealers, participate in an artificial market. Some human dealers may also participate in the artificial market with agents. All agents receive information relevant to a financial-price movement as input, and buy or sell virtual capital based on their rules. All agents can change their rules to improve performance in trading. In addition to agents, an artificial market has a market mechanism. The market mechanism is a way of accumulating supply and demand for participants and determining a price in the market. With agents and a market mechanism, an artificial market can simulate various market phenomena, such as financial bubbles.

The simulation of the artificial market we propose is outlined in Fig. 1. Our system consists of a market-mechanism module, agent interface, and Kaburobo agents.

**Fig. 1.** Framework of our artificial-market system.

3.1.1. Market-mechanism module

The market-mechanism module represents the core of artificial-market services. It collects orders to buy or sell from market participants, computer agents and/or human players. The collected orders are stored in this module's order database. Prices are continuously determined using the collected orders whenever selling prices (offer prices) and buying prices (bid prices) meet. A transaction (payment and exchange) is executed just after the price is determined. The determined price is stored in a price database. This module can provide market data to market participants on request. The market data contain price histories, trading volumes, and market orders.

3.1.2. Agent interface

Computer programs (agents) can participate in artificial markets. Our system uses the Financial-Information-eXchange (FIX) protocol to enable communication between agents and the market-mechanism module. Order information (the amount of the order, and the bid or offer price) and market data (e.g., price histories) are exchanged between them. FIX was originally a protocol for exchanging data in actual electronic trading between securities firms (sell side) and institutional investors (buy side) [11]. Many electronic-broking systems in actual financial markets use the FIX protocol. The FIX standard is open to the public and is determined by the FIX committee who consist of members of numerous financial institutions and system developers. By using the FIX protocol for communication inside an artificial-market service, agents in our system can easily be modified to participate in actual financial markets through electronic-broking systems.

Agents using the FIX protocol can directly connect to the artificial market. Agents without the FIX protocol can communicate orders and market data through an FIX wrapper in the agent interface. The FIX wrapper translates an agent's communication content into an FIX protocol, and vice versa. The agent interface stores orders that are received from agents, and it sends a list of orders to the market-mechanism module using the FIX protocol.

3.1.3. Kaburobo agents

Kaburobo agents are automated-trading strategies that can place orders using the Kaburobo Application Programming Interface (API) in the Kaburobo SDK. They can also participate in the Kaburobo competition and not only in this artificial market.

First, all Kaburobo agents receive data about past market prices and trading volumes from the market-mechanism module through the agent interface. Next, agents determine orders from past data according to their trading rules. The orders are sent to the agent interface using the Kaburobo API, and information on accumulated orders is sent to the market-mechanism module by the agent interface. The market-mechanism module determines the market price according to the received-order list. Then, all agents receive reports on the transaction results from the market-mechanism module through the agent interface. After all the transactions have been completed, the agent interface broadcasts that the session has closed to the agents. It then sends new market data to the agents, and broadcasts that a new session has started. The agents place new orders taking the new data into consideration.

4. Assessment of automated-trading strategies

It is difficult to evaluate automated-trading strategies objectively. The method usually used is a back test, where performance is measured by applying the trading strategy to past price data. However, this raises the possibility of overlearning from past data. It is necessary to evaluate strategies in various market environments to avoid overlearning. We therefore simulated various market environments using an artificial market and assessed automated-trading strategies.

4.1. Comparison of back test and artificial-market test

We prepared eight kinds of automated-trading strategies that are listed in Table 3 and given in the appendix. These trading strategies were common and basic in many financial markets [16,12]. Since three parameter sets were prepared for each kind of agent, there were 24 agents in total. They were evaluated with both a back test and an artificial market. Trading rules (initial assets and order methods) in our experiments were the same as those in the Kaburobo competition. As these agents were created using the Kaburobo SDK, they can also participate in the competition. There were some agents that actually participated in the competition.

First, we calculated the returns of the 24 agents using the actual price data of a certain stock on the Tokyo Stock Exchange in 2005 and 2006. The agents initially had cash of 50m yen (about 0.5 m dollars) but no stock, and could sell short.⁴ The rankings of the agents by the annual rate of return per year are listed in Table 4.

⁴ Short-selling was allowed in our tests like the Tokyo Stock Exchange. In some financial markets, however, short-selling is restricted. If the initial amounts of agents' stocks were equivalent and sufficient, agents do not need to sell short. The simulation results do not change on this condition.

Table 3
Parameter values of agents

Name	Parameter values
GoldenCross	{SMA, LMA} = {12 day, 24 day}
GoldenCrossShort	{SMA, LMA} = {5 day, 15 day}
GoldenCrossLong	{SMA, LMA} = {25 day, 75 day}
HLBand	$n = 10$
HLBandShort	$n = 5$
HLBandLong	$n = 25$
MACD	{SMA, LMA, PS} = {12 day, 26 day, 9 day}
MACDShort	{SMA, LMA, PS} = {6 day, 12 day, 5 day}
MACDLong	{SMA, LMA, PS} = {16 day, 40 day, 12 day}
Envelope	$P = 0.05$
EnvelopeNarrow	$P = 0.025$
EnvelopeWide	$P = 0.10$
PsychologicalLine	{PH, PL} = {0.75, 0.25}
PsychologicalLineNarrow	{PH, PL} = {0.60, 0.40}
PsychologicalLineWide	{PH, PL} = {0.85, 0.15}
RSI	{RH, RL} = {0.3, 0.7}
RSINarrow	{RH, RL} = {0.2, 0.8}
RSIWide	{RH, RL} = {0.4, 0.6}
TechnicalKun	$n = 20$
TechnicalKunShort	$n = 10$
TechnicalKunLong	$n = 100$
Bayesian	$TH = 0.6$
BayesianNarrow	$TH = 0.4$
BayesianWide	$TH = 0.8$

The details of rules and parameters are written in appendix.

Table 4
Results of back tests in 2005 and 2006.

2005			2006		
	Name	Return		Name	Return
1	GoldenCrossLong	12.92	1	PsychologicalLineWide	15.49
2	BayesianWide	6.35	2	TechnicalKunLong	14.32
3	HLBandShort	3.78	3	Envelope	12.61
4	PsychologicalLineNarrow	3.54	4	RSI	10.09
5	GoldenCrossShort	3.03	5	EnvelopeNarrow	9.71
6	GoldenCross	2.98	6	BayesianNarrow	7.91
7	Bayesian	2.17	7	BayesianWide	7.34
8	HLBand	0.46	8	RSIWide	6.54
9	BayesianNarrow	0.21	9	Bayesian	5.17
10	MACD	0.12	10	RSINarrow	2.89
11	MACDShort	0.08	11	EnvelopeWide	2.01
12	EnvelopeNarrow	0.06	12	GoldenCross	1.92
13	EnvelopeWide	0.00	13	PsychologicalLine	1.89
13	TechnicalKun	0.00	14	TechnicalKunShort	1.87
13	TechnicalKunShort	0.00	15	MACD	1.26
13	TechnicalKunLong	0.00	16	MACDShort	0.87
17	MACDLong	-0.35	17	TechnicalKun	0.08
18	HLBandLong	-0.62	18	PsychologicalLineNarrow	0.05
19	PsychologicalLineWide	-0.89	19	HLBandLong	-0.07
20	PsychologicalLine	-1.15	20	HLBand	-0.79
21	RSINarrow	-1.26	21	GoldenCrossLong	-1.16
22	RSIWide	-1.64	22	MACDLong	-1.23
23	RSI	-4.40	23	HLBandShort	-2.02
24	Envelope	-4.48	24	GoldenCrossShort	-5.64

The rankings were quite different for these two years. Their correlation coefficient was -0.462. If someone used automated-trading strategies ranked higher by the back test in 2005, they may have sustained quite heavy losses in 2006. Three agents in the top five in 2005, for example, sustained losses in 2006.

Next, we tested how these agents performed in the artificial market. The market participants were the 24 agents listed in Table 3 and three random trading agents. The random trading agents bought or sold with a probability of $\frac{1}{2}$. The initial conditions and order methods for the agents were the same as in the back test. But orders from agents were sent to the market-mechanism

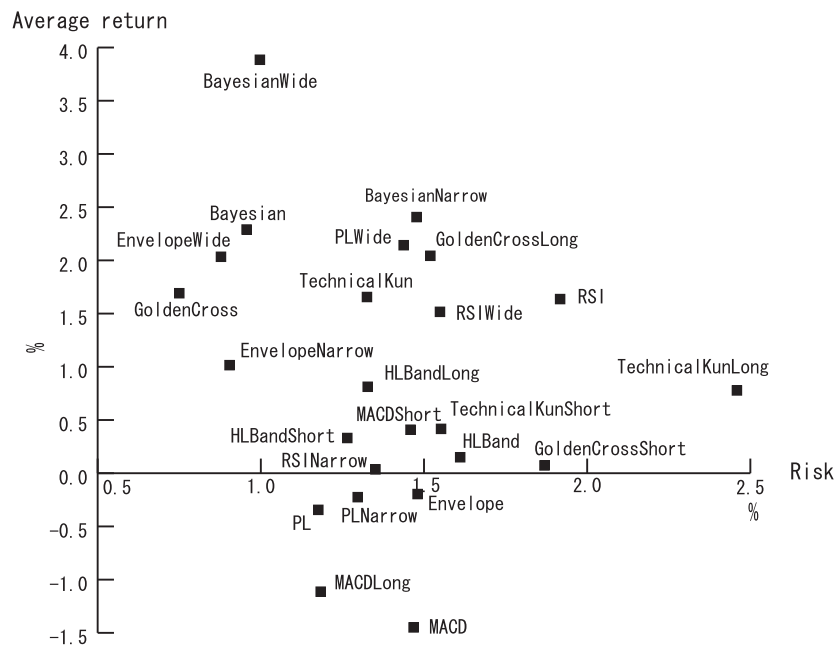


Fig. 2. Plots of risk (standard deviation of returns) and average returns of agents.

module, and this module determined the price in the artificial market. The price information is fed back to the agents. Orders of agents could change the market price in an artificial market test, unlike a back test.

We carried out 10 simulation runs with 100 time steps. As a result, the artificial market test was able to evaluate the overall feature of the agents' performance. First, we ranked the 24 agents by the average of final returns in 10 trials. The ranking achieved by simulation had a positive correlation with the rankings achieved by the back test both in 2005 (0.237) and 2006 (0.410), although the rankings in 2005 and 2006 had a negative correlation. Next, we examined the stability of agents' final returns. In a classical method, a risk of a trading strategy is measured by the volatility of short-term returns in one price path. However, it is well known that this measure may have a strong bias sometimes. Moreover, it cannot satisfy traders' demand because their main concern is not about a halfway change of returns but about a final profit [10]. Because the artificial market can generate an ensemble of price paths with their possibility, we can measure a risk of a trading strategy by standard deviation of *final returns* over simulation paths. Fig. 2 plots the average annual rate returns and standard deviation of returns over the 10 simulation runs. We can see which agents are high-risk strategies and which are low-risk. From the comparison with the back test both in 2005 and 2006, the artificial market test could evaluate risks of trading strategies well. For example, the GoldenCrossLong agent and PsychologicalLineWide agent had higher returns, but they had higher risks in the artificial market test (Fig. 2). This result was supported by the comparison between the back tests in 2005 and 2006. Although those agents were at the top in one year, they were ranked lower in the other year (Table 4). The BayesianWide agent has a lower risk, and it was stable with higher ranks both in the 2005 and 2006 back tests. In this experiment, only this type of agents were adaptive. They can change their rules during the simulation by a machine learning method. The higher performance of this type can imply possibilities of applying artificial intelligence technology to finance.⁵ These results demonstrate that tests in an

artificial market can be used to evaluate the returns and risks of automated-trading strategies.

5. Does automated trading cause fluctuations in market?

Over 20% of all orders in 2006 in foreign-exchange markets were entered by algorithmic traders that were supported by computer programs [24]. Other markets such as equity markets seem to have a higher proportion of automated trades and the proportion is growing steadily. As automated trading continues to spread, its impact on markets cannot be ignored. It is especially important to know whether the spread of certain types of trading strategies will threaten the stability of financial markets.

We can test what influence trading strategies have under various conditions before they are actually used, by simulating an artificial market with automated-trading strategies. The results obtained may suggest a plan of operation for the tested trading strategies taking their market impact into consideration. For example, if an automatic-trading strategy is making the market unstable, a trader can stop it or changes its parameters. Consequently, excessive fluctuations in the market that the trading strategy is causing can be prevented in advance. Traders can reduce risks caused by fluctuations.

5.1. Test of impact on market of automated-trading strategies

We examined how the way of combination of trading strategies affected the market stability. If the trading strategies in Table 3 were directly used in this test, the number of combination would become huge. Then, we tested the market impact of two basic trading algorithms, i.e., momentum trading (trade following trends) and contrary trading (trade against trends). If there is an upward (downward) trend, a momentum-trading algorithm places a buying (selling) order since it has forecast that the trend will continue. A contrary-trading algorithm will sell (buy) since it has forecast that the previous upward (downward) trend will reverse. All the strategies in Table 3 can be classified into one of these two simplified algorithms. We constructed an artificial market with 55 agents and one stock.

⁵ If other agents are also adaptive using other machine learning method, the results may change. Thus, the further examination are required.

Table 5
Agent rules.

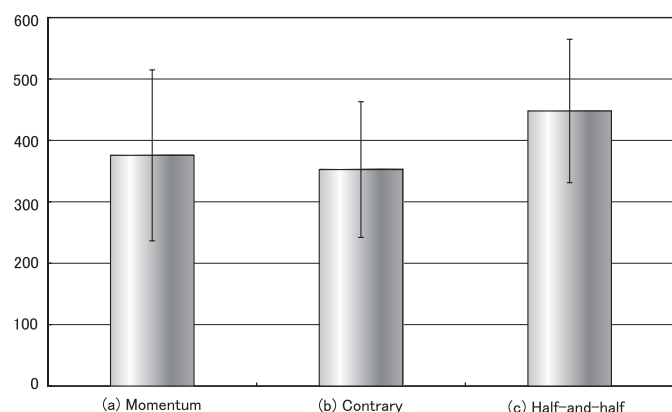
Condition	Action
<i>Momentum trading agent</i>	
Upward previous trend →	Buy 1000 units at $(1 + \alpha_t) \times P_{t-1}$
Downward previous trend →	Sell 1000 units at $(1 + \alpha_t) \times P_{t-1}$
<i>Contrary trading agent</i>	
Upward previous trend →	Sell 1000 units at $(1 + \alpha_t) \times P_{t-1}$
Downward previous trend →	Buy 1000 units at $(1 + \alpha_t) \times P_{t-1}$
<i>Random trading agent</i>	
Probability of $\frac{1}{2}$ →	Buy 1000 units at $(1 + \alpha_t) \times P_{t-1}$
Probability of $\frac{1}{2}$ →	Sell 1000 units at $(1 + \alpha_t) \times P_{t-1}$

P_{t-1} is a price at the previous session.

α_t is a random value between -0.1 and 0.1 .

Table 6
Combination of agent types.

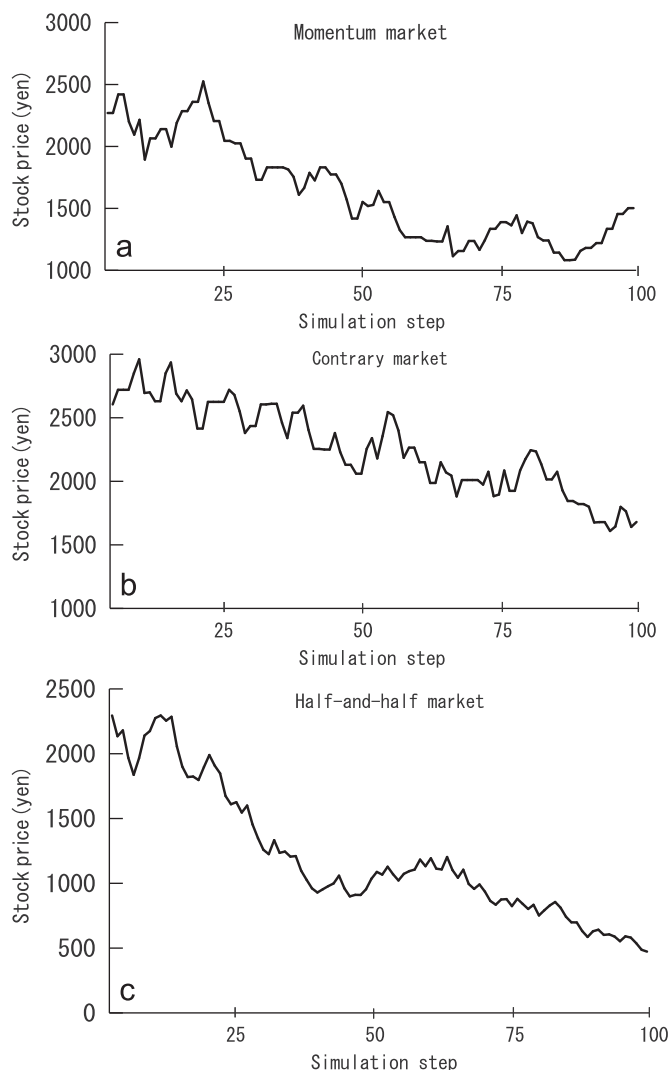
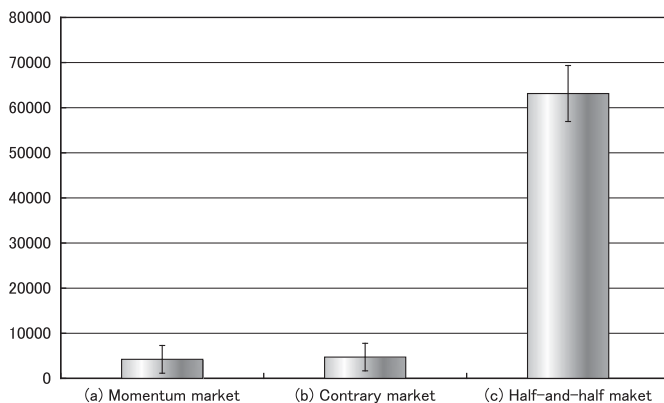
	Momentum agent	Contrary agent	Random agent
Momentum market	50	0	5
Contrary market	0	50	5
Half-and-half market	25	25	5

**Fig. 3.** Comparison of market impact: standard deviation of prices (average value over 10 simulation runs).

There were three kinds of agents participating in the artificial market, i.e., momentum-trading, contrary-trading, and random-trading agents (see Table 5). Random-trading agents sell or buy with a probability of $\frac{1}{2}$.

Three kinds of combinations of agents as listed in Table 6 were prepared, and variations in the market price in each artificial market were compared. We carried out 10 simulation runs with 100 time steps. As a result, in the half-and-half market with both contrary-trading and momentum-trading algorithms, as seen in Fig. 3, the market became the least stable.

We investigated typical paths with price fluctuations close to the average for each market in Table 6 to find the reason for this least stability. There were almost no trends in the contrary market and the price had settled within a certain range (Fig. 4b). This is because orders by contrary agents tended to erase or reverse trends immediately. A trend appeared in the momentum market (Fig. 4a). This is because orders by momentum agents supported the trend. However, the trend in the momentum market was shorter than that in the half-and-half market. The period trends lasted in the half-and-half market and market fluctuations increased (Fig. 4c). In order to examine a reason of these differences, we compared the trading volumes between the three markets (Fig. 5). The volume for the half-and-half market

**Fig. 4.** Comparison of price fluctuation of typical paths.**Fig. 5.** Comparison of average trading volume of typical paths.

was clearly larger than those for the other markets. Since there were always plenty of orders to both buy and sell in the half-and-half market, the trends lasted longer. However, since orders occasionally converged to one side in the momentum market, trends did not continue. Thus, our results revealed that the market impact of automated-trading strategies may not only depend on their rule content but also on the way they are combined with other strategies.

6. Conclusion

The objective of this paper was to expand the artificial-market approach to the practical application in financial markets. We constructed an artificial-market system with an interface with the automated-trading strategies. Then, using an artificial-market simulation, we conducted the two types of evaluations of automated-trading strategies that could participate in a Kaburobo competition. (1) Our system evaluated the risks and returns of the strategies in various market environments. An evaluation using the artificial market was able to provide better information than a conventional evaluation using a back test. (2) Our system could also test the market impact of automated-trading strategies. Our results revealed that the market impact of the strategies may not only depend on their rule content but also on the way they are combined with other strategies.

For the practical application of the artificial-market system, the following future works will be needed. First, it is necessary to introduce an agent that estimates a proper price level from the fundamental information such as the performance of a company and various economic conditions. Thereby, we will be able to evaluate a trading strategy in consideration of external conditions outside a market. At present, all the agents only used the past price information, and the influence of economic fundamentals was not taken into consideration at all. The second points is to develop various types of adaptive agents with artificial intelligence methods. In this study, only the Bayesian agent was adaptive. If the variety of adaptive agents is introduced, we will be able to examine the phase transition of a market environment, like the dynamic game theory. Finally, we can test some market regulations such as up-tick rule and a circuit breaker, using our system. The effect of a regulation will be tested in advance, by the comparison between simulation results with the regulation and those without it.

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Appendix A. Types of agents

Trading strategies of the agents in Table 3 are as follows.

Variables: $p(t)$ is a price at day t ; $v(t)$ is a trading volume at day t ; $rand(n)$ is a random value from 0 to $n - 1$.

Methods: Trading methods are as follows.

```

(1) buy()
    BuyPrice :=  $p(t) \pm rand(p(t-1) - p(t))/2$ 
    Buy 1 unit at BuyPrice
sell()
    SellPrice :=  $p(t) \pm rand(p(t-1) - p(t))/2$ 
    Sell 1 unit at SellPrice

(2) GoldenCross Agent
     $MA_L(t)$  := Moving Average from  $t - LMA$  to  $t - 1$ 
     $MA_S(t)$  := Moving Average from  $t - SMA$  to  $t - 1$ 
    if  $MA_S(t-1) < MA_L(t-1)$  and  $MA_L(t) > MA_S(t)$  then
        buy()
    else if  $MA_S(t-1) > MA_L(t-1)$  and  $MA_L(t) < MA_S(t)$  then
        sell()
    end if

(3) HLBand Agent
    HB :=  $\max(p(t-1), \dots, p(t-n))$ 
    LB :=  $\min(p(t-1), \dots, p(t-n))$ 
    if  $p(t) > HB$  then
        buy()
    else if  $p(t) < LB$  then
        sell()
    end if

```

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(4) MACD Agent
     $EMA_S(t)$  := Exponential Moving Average from  $t - SMA$  to  $t - 1$ 
     $EMA_L(t)$  := Exponential Moving Average from  $t - LMA$  to  $t - 1$ 
     $MACD(t) = EMA_S(t) - EMA_L(t)$ 
     $Signal(t) = \frac{1}{PS_{n=1,PS}MACD(t-n)}$ 
    if  $MACD(t-1) < Signal(t-1)$  and  $MACD(t) > Signal(t)$  then
        buy()
    else if  $MACD(t-1) > Signal(t-1)$  and  $MACD(t) < Signal(t)$  then
        sell()
    end if

(5) Envelope Agent
     $MA(t)$  := Moving Average from  $t - n$  to  $t - 1$ 
    if  $p(t) < MA(t) \cdot (1 - P)$  then
        buy()
    else if  $p(t) > MA(t) \cdot (1 + P)$  then
        sell()
    end if

(6) PsychologicalLine Agent
    gain := 0
    i := 0
    while  $i < 12$  do
        if  $p(t-i) > p(t-(i+1))$  then
            gain := gain + 1
        end if
        i := i + 1
    end while
    if gain/12 < PL then
        buy()
    else if gain/12 > PH then
        sell()
    end if

(7) RSI Agent
    gain := 0
    loss := 0
    i := 0
    while  $i < 12$  do
         $dp := p(t-i) - p(t-(i+1))$ 
        if  $dp > 0$  then
            gain := gain + 1
        else if  $dp < 0$  then
            loss := loss + 1
        end if
        i := i + 1
    end while
    if gain/(gain + loss) < PL then
        buy()
    else if gain/(gain + loss) > PH then
        sell()
    end if

(8) Mixture of Technical Analysis (TechnicalKun)
    signal := 0
    //Check envelope signal
     $MA(t)$  := Moving Average from  $t - n$  to  $t - 1$ 
    if  $p(t) < MA(t) \cdot (1 - P)$  then
        signal := signal + 1
    else if  $p(t) > MA(t) \cdot (1 + P)$  then
        signal := signal - 1
    end if
    //Check RSI signal
    gain := 0, loss := 0, i := 0
    while  $i < 12$  do
         $dp := p(t-i) - p(t-(i+1))$ 
        if  $dp > 0$  then
            gain := gain + 1
        else if  $dp < 0$  then
            loss := loss + 1
        end if
        i := i + 1
    end while
    if gain/(gain + loss) < PL then
        signal := signal + 1
    else if gain/(gain + loss) > PH then
        signal := signal - 1
    end if
    //Check VolumeRatio signal
    gain := 0, loss := 0, flat := 0, i := 0
    while  $i < 12$  do
         $dp := p(t-i) - p(t-(i+1))$ 

```

```

if  $dp > 0$  then
   $gain := gain + v(t - i)$ 
else if  $dp < 0$  then
   $loss := loss + v(t - i)$ 
else
   $flat := flat + v(t - i)$ 
end if
 $i := i + 1$ 
end while
 $VR = \frac{gain + flat/2}{loss + flat/2}$ 
if  $VR < 0.7$  then
   $signal := signal + 1$ 
else if  $VR < 4.5$  then
   $signal := signal - 1$ 
end if
//Check final signal
if  $signal = 3$  then
  buy()
else if  $signal = -3$  then
  sell()
end if
if Have Long-Position more than  $n$  days then
  sell()
else if Have Short-Position more than  $n$  days then
  buy()
end if
Bayesian Agent
 $i = LearnDays + TestDays$ 
while  $i > TestDays$  do
   $p_i = \{p(t - i), p(t - (i + 1)), \dots, p(t - (i + n))\}$ 
   $GainRate := \frac{\max(p(t - (i - 1)), p(t - (i - 2)), \dots, p(t - (i - TestDays)))}{p(t - (i - 1))}$ 
   $LossRate := \frac{\min(p(t - (i - 1)), p(t - (i - 2)), \dots, p(t - (i - TestDays)))}{p(t - (i - 1))}$ 
  if  $GainRate > 1.05$  then
    Update GainTrendGroup by  $p_i$  by using Bayes theory
  else if  $LossRate < 0.95$  then
    Update LossTrendGroup by  $p_i$  by using Bayes theory
  end if
end while
 $p = \{p(t - 1), p(t - 2), \dots, p(t - n)\}$ 
 $gain := 0$ 
 $loss := 0$ 
for all  $p_i$  such that  $p_i \in GainTrendGroup$  do
  if  $Correlation(p_i, p) > TH$  then
    buy()
  return
end if
end for
for all  $p_i$  such that  $p_i \in LossTrendGroup$  do
  if  $Correlation(p_i, p) > TH$  then
    sell()
  return
end if
end for

```

(8)

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